

# Xingyu Bruce Liu

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## Current Position

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2026 – now    📖 **Adobe Research** Research Scientist

## Education

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- 2020 – 2026    📖 **University of California, Los Angeles**  
Ph.D. Candidate, UCLA HCI Lab  
Advised by Professor Xiang ‘Anthony’ Chen  
Thesis: *Beyond Taking Turns in Human-AI Interaction*
- 2020 – 2022    📖 **University of California, Los Angeles**  
M.S. Electrical and Computer Engineering  
Advised by Professor Xiang ‘Anthony’ Chen  
Distinguished Master’s Thesis Research Award, UCLA ECE
- 2023 summer    📖 **The University of Tokyo**  
Visiting Ph.D. Student, Computer Science, Igarashi Lab  
Advised by Professor Takeo Igarashi
- 2016 – 2020    📖 **Carnegie Mellon University**  
B.S. Statistics and Machine Learning, Human-Computer Interaction  
Minor in Computer Science  
with University Honors

## Publications

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### Peer-reviewed Publications

- [1] **Xingyu Bruce Liu**, Mira Dontcheva, and Dingzeyu Li. 2026. A Text-Native Interface for Generative Video Authoring. In *Submission* (Under Review).
- [2] **Xingyu Liu**. 2025. Thought as a Substrate in Human-AI Interaction. In *UIST 2025 Doctoral Symposium* (UIST ’25). Association for Computing Machinery, Busan, Republic of Korea. 📄 DOI: 10.1145/3746058.3758466.
- [3] Ruolin Wang, **Xingyu Liu**, Biao Wang, Wayne Zhang, Ziqian Liao, Ziwen Li, Amy Pavel, and Xiang ‘Anthony’ Chen. 2025. CoSight: Exploring Viewer Contributions to Online Video Accessibility Through Descriptive Commenting. In *Proceedings of the 2025 ACM Symposium on User Interface Software and Technology* (UIST ’25). Association for Computing Machinery, Busan, Republic of Korea. 📄 DOI: 10.1145/3746059.3747747.
- [4] **Xingyu Bruce Liu**, Shitao Fang, Weiyan Shi, Chien-Sheng Wu, Takeo Igarashi, and Xiang ‘Anthony’ Chen. 2025. Proactive Conversational Agents with Inner Thoughts. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI ’25). Association for Computing Machinery, Yokohama, Japan. 📄 DOI: 10.1145/3706598.3713760.

- [5] **Xingyu Bruce Liu**, Jiahao Nick Li, David Kim, Xiang 'Anthony' Chen, and Ruofei Du. 2024. Human I/O: Towards a Unified Approach to Detecting Situational Impairments. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (CHI '24). ACM.  DOI: 10.1145/3613904.3642065.  **Best Paper Honorable Mention.**
- [6] Ruofei Du, Na Li, Jing Jin, Michelle Carney, Scott Miles, Maria Kleiner, Xiuxiu Yuan, Yinda Zhang, Anuva Kulkarni, **Xingyu Bruce Liu**, Ahmed Sabie, Sergio Escolano, Abhishek Kar, Ping Yu, Ram Iyengar, Adarsh Kowdle, and Alex Olwal. 2023. Rapsai: Accelerating Machine Learning Prototyping of Multimedia Applications Through Visual Programming. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). ACM.  DOI: 10.1145/3544548.3581338.  **Best Paper Honorable Mention.**
- [7] **Xingyu Bruce Liu**, Vladimir Kirilyuk, Xiuxiu Yuan, Alex Olwal, Peggy Chi, Xiang 'Anthony' Chen, and Ruofei Du. 2023. Visual Captions: Augmenting Verbal Communication With On-the-Fly Visuals. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). ACM.  DOI: 10.1145/3544548.3581566.
- [8] **Xingyu Bruce Liu\***, Joanne Leong\*, Yuanyang Teng\*, Hanseul Jun, Sven Kratz, Yu Jiang Tham, Andrés Monroy-Hernández, Brian A. Smith, and Rajan Vaish. 2023. Social Wormholes: Exploring Preferences and Opportunities for Distributed and Physically-Grounded Social Connections. In *Proceedings of the 26th ACM Conference On Computer-Supported Cooperative Work And Social Computing* (CSCW '23). ACM.
- [9] **Xingyu Bruce Liu**, Ruolin Wang, Dingzeyu Li, Xiang 'Anthony' Chen, and Amy Pavel. 2022. CrossAny: Identifying Video Accessibility Issues via Cross-Modal Grounding. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (UIST '22). ACM, Bend, OR, USA.  DOI: 10.1145/3526113.3545703.  **Best Paper Award.**
- [10] **Xingyu Liu**, Patrick Carrington, Xiang 'Anthony' Chen, and Amy Pavel. 2021. What Makes Videos Accessible to Blind and Visually Impaired People? In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (CHI '21). ACM, Yokohama, Japan.  DOI: 10.1145/3411764.3445233.
- [11] Cole Gleason, Amy Pavel, **Xingyu Liu**, Patrick Carrington, Lydia B. Chilton, and Jeffrey P. Bigham. 2019. Making Memes Accessible. In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '19). ACM, Pittsburgh, PA, USA.  DOI: 10.1145/3308561.3353792.

## Late-Breaking Works, Posters, Demos

- [12] **Xingyu Bruce Liu**, Vladimir Kirilyuk, Xiuxiu Yuan, Peggy Chi, Alex Olwal, Xiang 'Anthony' Chen, and Ruofei Du. 2023. Experiencing Visual Captions: Augmented Communication with Real-time Visuals using Large Language Models. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (UIST '23 Adjunct). ACM, New York, NY, USA.  DOI: 10.1145/3586182.3615978.
- [13] **Xingyu Bruce Liu**, Jun Zhang, Leonardo Ferrer, Susan Xu, Vikas Bahirwani, Boris Smus, Alex Olwal, and Ruofei Du. 2023. Modeling and Improving Text Stability in Live Captions. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI EA '23). ACM.  DOI: 10.1145/3544549.3585609.

## Patents

- [14] Hanseul Jun, Sven Kratz, Joanne Leong, **Xingyu Liu**, Andrés Monroy-Hernández, Brian Anthony Smith, Yu Jiang Tham, and Rajan Vaish. 2025. Snapshot messages for indicating user state. Patent No. US12363419. (July 2025).
- [15] Ruofei Du and **Xingyu Liu**. 2024. Determining Communication Channel Based on Limitation of User. Application No. US20240412495. (December 2024).

- [16] Ruofei Du and **Xingyu Liu**. 2024. Generating an Avatar Expression. Application No. US20240265605. (August 2024).
- [17] Ruofei Du, Alex Olwal, and **Xingyu Liu**. 2024. System and Method for Generating Visual Captions. Application No. US20240330362. (October 2024).
- [18] Hanseul Jun, Sven Kratz, Joanne Leong, **Xingyu Liu**, Andrés Monroy-Hernández, Brian Anthony Smith, Yu Jiang Tham, and Rajan Vaish. 2024. Scan-based messaging for electronic eyewear devices. Patent No. US12149490. (November 2024).
- [19] Hanseul Jun, Sven Kratz, Joanne Leong, **Xingyu Liu**, Andrés Monroy-Hernández, Brian Anthony Smith, Yu Jiang Tham, and Rajan Vaish. 2024. Social connection through distributed and connected real-world objects. Patent No. US12072489. (August 2024).
- [20] Bing Liu and **Xingyu Liu**. 2021. Method, device and computer product for predicting disk failure. Patent No. US10996861. (May 2021).
- [21] Bing Liu and **Xingyu Liu**. 2021. Method, device, and computer program product for facilitating prediction of disk failure. Patent No. US11036572. (June 2021).

## Awards and Honors

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- 2025      **Dissertation Year Award**, UCLA
- 2024      **Best Paper Honorable Mention** (top 5%), CHI 2024
- 2023      **Amazon Ph.D. Fellowship**, 2023
-  **Best Paper Honorable Mention** (top 5%), CHI 2023
-  **ED Rice Outstanding Master Student Award**, UCLA Engineering School
- 2022      **Best Paper Award** (top 3), UIST 2022
-  **Distinguished Master's Thesis Research Award**, UCLA ECE Department.
- 2020-2022      **Departmental Fellowship**, UCLA ECE Department, \$65,000.
- 2018      **Best Social Impact Award**, TartanHacks (40+ teams).
- 2017      **First Place, Most Technical Award**, HackNY (20+ teams).
- 2016      **Mizuho Scholar**, Mizuho & Wing Hang Bank Scholarship and Charity Funds.
- 2016 – 2020      **Dean's List**, Carnegie Mellon University.

## Professional Experience

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- 2025 summer      **Adobe Research**, Research Scientist Intern.  
Novel interface for generative video authoring.  
Advised by Dr. Dingzeyu Li and Dr. Mira Dontcheva.
- 2024 summer      **Meta Reality Labs**, Research Scientist Intern.  
Novel human-LLM interaction paradigm.  
Advised by Dr. Mark Parent and Dr. Ben Lafreniere.
- 2022 spring/summer      **Google**, Student Researcher.  
Augmented language and contextual computing.  
Four papers published at CHI and UIST.  
Advised by Dr. Ruofei Du.

## Professional Experience (continued)

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- 2021 summer      **Snap Research**, Research Intern.  
AR-based physical connections for remote awareness between friends.  
Paper published at CSCW.  
Advised by Dr. Rajan Vaish and Dr. Brian A. Smith.

## Service

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- 2020 – Now      **Associate Chair**  
CHI 2025, 2026
-  **Video Presentation Chair**  
CHI 2025
-  **Reviewer**  
CHI 2021-2026, UIST 2020-2025, CSCW 2020, 2021, 2025, ICML 2023, IMWUT 2023
-  **Special Recognitions as a Reviewer**  
CHI 2022, CHI 2023 x 2, CHI 2024, UIST 2024, IMWUT 2023